MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $P_A(B)$ has as denominator
   - A. $P(B)$
   - B. $P(A)$
   - C. $P(A \cap B)$
   - D. $1 - P(A)$
2. $P(A) = 0.4$, $P_A(B) = 0.2$: intersection
   - A. 0.08
   - B. 0.6
   - C. 0.5
   - D. 0.2
3. A partition consists of events that are
   - A. equal
   - B. all independent
   - C. equally probable
   - D. disjoint and covering the sample space
4. Independence is equivalent to
   - A. $P(A) = P(B)$
   - B. $P(A \cap B) = 0$
   - C. $P(A \cap B) = P(A)P(B)$
   - D. $A = B$
5. Two disjoint events of positive probability are
   - A. independent
   - B. not independent
   - C. equal
   - D. necessarily complementary

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. Three independent tosses of a fair coin, at least one head:
   - A. $\frac{1}{8}$
   - B. $\frac{3}{8}$
   - C. $\frac{7}{8}$
   - D. $\frac{1}{2}$
2. Two independent trials with $p = \frac{1}{3}$, two successes:

A. $\frac{1}{9}$
B. $\frac{2}{3}$
C. $\frac{1}{3}$
D. $\frac{4}{9}$

3. The branches leaving the same node sum to
   A. 0
   B. 2
   C. the probability of the path
   D. 1
4. A simulation gives a frequency equal to 0.51. This
   A. proves $p = 0.51$
   B. is a fluctuating observation
   C. proves $p = \frac{1}{2}$
   D. refutes $p = \frac{1}{2}$
5. Drawing without replacement, after a red ball from 3 red and 2 blue, the probability of red is:
   A. $\frac{3}{5}$
   B. $\frac{3}{4}$
   C. $\frac{1}{2}$
   D. $\frac{2}{5}$

True or false? Justify.

1. $P_A(B) = P_B(A)$ for all events of positive probability.
2. If A and B are independent, so are A and $\overline{B}$.
3. Two successive draws are always independent.

Self-assessment

I identify the right denominator; I check the sums at the nodes; I justify independence; I count the paths without confusing them.
For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.